



Original Contribution

Glycoumarin ameliorates alcohol-induced hepatotoxicity via activation of Nrf2 and autophagy

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ABSTRACT

Licorice, a traditional Chinese medicine, has been used to treat various diseases, including liver disease, for centuries. However, the chemical basis and biological mechanisms underlying the biological functions of licorice remain elusive. The purpose of the current study was to test the hepatoprotective effect of glycoumarin (GCM), a representative coumarin in licorice, using animal models of both chronic and acute alcoholic liver injury. C57BL/6J mice were used to evaluate the hepatoprotective effect of GCM on liver injury induced by either chronic or acute ethanol exposure. AML-12 and HepG2 cells were utilized to determine the functional role of Nrf2 in the hepatoprotective effect of GCM and to decipher the mechanisms of GCM-induced Nrf2 activation. We found that treatment with GCM leads to a significant reduction in hepatotoxicity in response to either chronic or acute ethanol exposure. Further mechanistic investigations reveal that activation of Nrf2 via the p38 pathway and induction of autophagy by GCM contribute to its hepatoprotective activity. In addition, we demonstrate that p62 upregulation by a transcriptional mechanism also contributes to Nrf2 activation via a positive feedback loop. Our study has identified GCM as a novel active ingredient that contributes to the hepatoprotective activity of licorice.

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Alcoholic liver disease (ALD)¹ is a major cause of liver-related mortality not only in developed but also in developing countries. Epidemiologic study reveals that the incidence of alcoholic liver disease in China is rapidly increasing [1]. Development of novel agents that can improve the efficacy of ALD prevention and treatment is urgently needed.

Herbal medicine as an alternative approach has been used in the treatment of liver diseases for a long time [2,3]. However, scientific validation of its efficacy is apparently needed. Licorice, a popular edible and medicinal plant, has been widely used to treat various diseases, including infectious and inflammatory disease, cancer, spasmodic disease, and liver disease [4,5]. A number of

licorice-derived compounds have been reported to be associated with these pharmacological activities [5–7], yet the chemical basis and biological mechanisms underlying these biological functions of licorice have not been well elucidated. Glycoumarin (GCM) is a representative coumarin in licorice with favorable bioavailability features in vivo [8]. It has been shown that GCM possesses antiviral [9,10], anti-inflammatory [11], and antispasmodic effects [12]. We hypothesized that GCM could be one active component of licorice that contributes to its liver protection activity. We therefore carried out such investigation using animal models of both chronic and acute alcoholic liver injury. Our results show that GCM exhibits a strongly preventive effect on alcohol-induced hepatotoxicity in the in vivo mouse models of both chronic and acute alcoholic liver injury. Further mechanistic studies reveal that the hepatoprotective effect of GCM is attributed to activation of nuclear factor erythroid 2-related factor 2 (Nrf2) and autophagy. To our knowledge, this is the first report showing that GCM is a potent hepatoprotective agent isolated from licorice against alcoholic liver disease. Our findings provide new scientific evidence for licorice as an herbal remedy for alcoholic liver disease.

Abbreviations: ALD, alcoholic liver disease; GCM, glycoumarin; TG, triglyceride; ALT, alanine aminotransferase; AST, aspartate aminotransferase; Nrf2, nuclear factor erythroid 2-related factor 2; HO-1, heme oxygenase 1; GCLC, glutamate-cysteine ligase catalytic subunit; DAPI, 4',6'-diamidino-2-phenylindole; H&E, hematoxylin and eosin; SnMP, stannic mesoporphyrin; LC3, microtubule-associated protein 1 light chain 3; BAF, bafilomycin A1; Keap1, Kelch-like ECH-associated protein 1

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1. Materials and methods

1.1. Chemicals and reagents

GCM (purity 98% or higher) was isolated from the rhizomes of glycyrrhiza by the Ye lab as reported previously [4]. 4',6'-Diamidino-2-phenylindole (DAPI), oil red O solution, and bafilomycin A1 were purchased from Sigma Chemical Co. (St. Louis, MO, USA). Antibody for microtubule-associated protein 1 light chain 3 (LC3) was purchased from MBL International Corp. Antibodies specific for p62, Nrf2, HO-1, Keap1, phospho-p38, phospho-JNK1/2, phospho-ERK1/2, and β -actin were purchased from Cell Signaling Technology (Beverly, MA, USA). Antibody for GCLC was purchased from Abcam (Cambridge, MA, USA). Nrf2 small interfering RNA (siRNA), p62 siRNA, and nontargeting siRNA were purchased from Santa Cruz Biotechnology (Santa Cruz, CA, USA). RNeasy Minikit and QuantiFast Probe RT-PCR Kit were purchased from Qiagen (Valencia, CA, USA). High Capacity cDNA Reverse Transcription Kit (4368814) was purchased from Invitrogen (Carlsbad, CA, USA).

1.2. Cell culture and treatments

AML-12 and HepG2 cell lines were obtained from the American Type Culture Collection. The cells were grown in Dulbecco's modification of Eagle's medium supplemented with 10% fetal bovine serum without antibiotics. At 24–48 h after plating, when the cells were 50–60% confluent, the medium was changed before starting the treatment with GCM and/or other agents.

1.3. Western blotting

Cell lysates were prepared in ice-cold radio-immunoprecipitation assay buffer (20 mM Tris-HCl (pH 7.5), 150 mM NaCl, 1 mM Na₂EDTA, 1 mM EGTA, 1% NP-40, 1% sodium deoxycholate, 2.5 mM sodium pyrophosphate, 1 mM β -glycerophosphate, 1 mM Na₃VO₄, 1 μ g/ml leupeptin). Cell lysate proteins were resolved by electrophoresis and transferred to a polyvinylidene fluoride membrane (Millipore). The blot was then probed with a primary antibody followed by incubation with the appropriate horseradish peroxidase-conjugated secondary antibodies. The signal was visualized by enhanced chemiluminescence (Fisher/Pierce) and recorded on an X-ray film (Eastman Kodak Co.).

1.4. Reverse transcription-PCR

Total RNA was extracted at 24 h after GCM treatment using an RNeasy Minikit (Qiagen) and was used for reverse transcription according to the manufacturer's instructions (Invitrogen). The primers used for PCR analysis were synthesized by Sangon Biotech (Shanghai, China). The sequences of the primers were as follows: p62, 5'-AAGCCGGGTGGGAATGTTG-3' (forward) and 5'-GCTTGGCCTTCGGATTCT-3' (reverse), and glyceraldehyde-3-phosphate dehydrogenase (GAPDH), 5'-CAGCCTCAAGATCATCAGCA-3' (forward) and 5'-GTCTTCTGGGTGGCAGTGAT-3' (reverse). The quantification of real-time PCR products was performed using a QuantiFast Probe RT-PCR Kit (Qiagen). The housekeeping gene GAPDH was used as an internal control for RNA integrity and loading normalization. The relative amount of target mRNA was calculated by the comparative cycle threshold (C_t) method ($2^{-\Delta\Delta C_t}$) by normalizing target mRNA C_t values to those for GAPDH.

1.5. Immunofluorescence assay for Nrf2 localization

HepG2 cells were treated with GCM for 24 h. After the treatment, the cells were fixed with 4% paraformaldehyde for 15 min at

room temperature and permeabilized with 0.25% Triton X-100 in phosphate-buffered saline (PBS) at room temperature for 10 min and then blocked with 1% bovine serum albumin for 30 min at room temperature. After being blocked, the samples were incubated with anti-Nrf2 antibody at room temperature for 2 h. The cells were then washed and incubated with secondary fluorescein isothiocyanate-labeled goat anti-mouse antibody (MBL, Nagoya, Japan) for 1 h at room temperature. After three washes with PBS, the samples were counterstained with DAPI and examined by confocal microscopy.

1.6. RNA interference

The cells were transfected with 50 nmol/L Nrf2 siRNA or p62 siRNA or nontargeting siRNA using the INTERFERin siRNA transfection reagent according to the manufacturer's instructions (Polyplus-Transfection, Inc., New York, NY, USA) for 36 h and then were used for subsequent experiments.

1.7. Autophagy detection

Autophagy induction was determined by two methods. The first method was Western blotting analysis for conversion of the LC3-I to LC3-II. The second one was immunofluorescence staining for LC3 localization.

1.8. Animals and treatments

Eight- to 10-week-old C57BL/6J mice were obtained from The Jackson Laboratory (Bar Harbor, ME, USA). Animal care and procedures were approved by the institutional animal care and use committee. For the chronic alcoholic liver injury model [13], the mice were randomly distributed into six groups: control, ethanol, GCM 10 mg/kg, GCM 20 mg/kg, GCM 10 mg/kg + ethanol, and GCM 20 mg/kg + ethanol. Each group consisted of six mice. The ethanol group was fed the Lieber–DeCarli ethanol diet (Trophic Animal Feed, Nantong, China), with increasing 1–5% (wt/vol) ethanol for the first 7 days and then 5% (wt/vol) ethanol for 4 weeks followed by gavage of a single dose of ethanol (5 g/kg). The control group was fed the pair-fed control diet. GCM was given once a day (ip) for the last 3 weeks of the experiment. The animals were sacrificed 8 h after the last binge ethanol administration. Samples including blood and liver tissue were collected. Portions of liver tissue were fixed in neutral-buffered formalin, whereas others were frozen immediately in liquid nitrogen and stored at -80°C , and plasma was stored at -80°C for further analysis.

For the acute alcoholic liver injury model, 8- to 10-week-old mice were administered alcohol by gavage at a dose of 6 g/kg bw two times at 12 h intervals [14]. To determine the preventive effect of GCM on acute alcoholic liver damage, GCM or vehicle was given daily (20 and 30 mg/kg, ip) for 3 days prior to ethanol administration. Each group consisted of six mice. The mice were sacrificed 9 h after the second dose of ethanol and blood samples were collected for the measurement of alanine aminotransferase (ALT)/aspartate aminotransferase (AST).

1.9. Histopathology assessment

Neutral-buffered formalin-fixed liver samples were processed to paraffin sections and stained with hematoxylin and eosin (H&E). Histopathology evaluation was performed by a pathologist (Dr. Fei Pei, a pathologist in the Department of Pathology at Peking University) and scored as follows: steatosis < 5%, 0; 5–30%, 1; 31–50%, 2; 51–70%, 3; > 75%, 4 [15]. Frozen tissue sections were stained with oil red O for analysis of lipid storage in the liver.

1.10. Biochemical assay

Serum ALT and AST activity was measured using a commercially available kit (Biosino Bio-Technology, Beijing, China). Hepatic triglyceride content was determined using an EnzyChrom triglyceride assay kit (BioAssay Systems, Hayward, CA, USA).

1.11. Statistical analysis

Data are presented as the mean $\pm$ SD. These data were analyzed by ANOVA with appropriate post hoc comparisons among means. $P < 0.05$ was considered statistically significant.

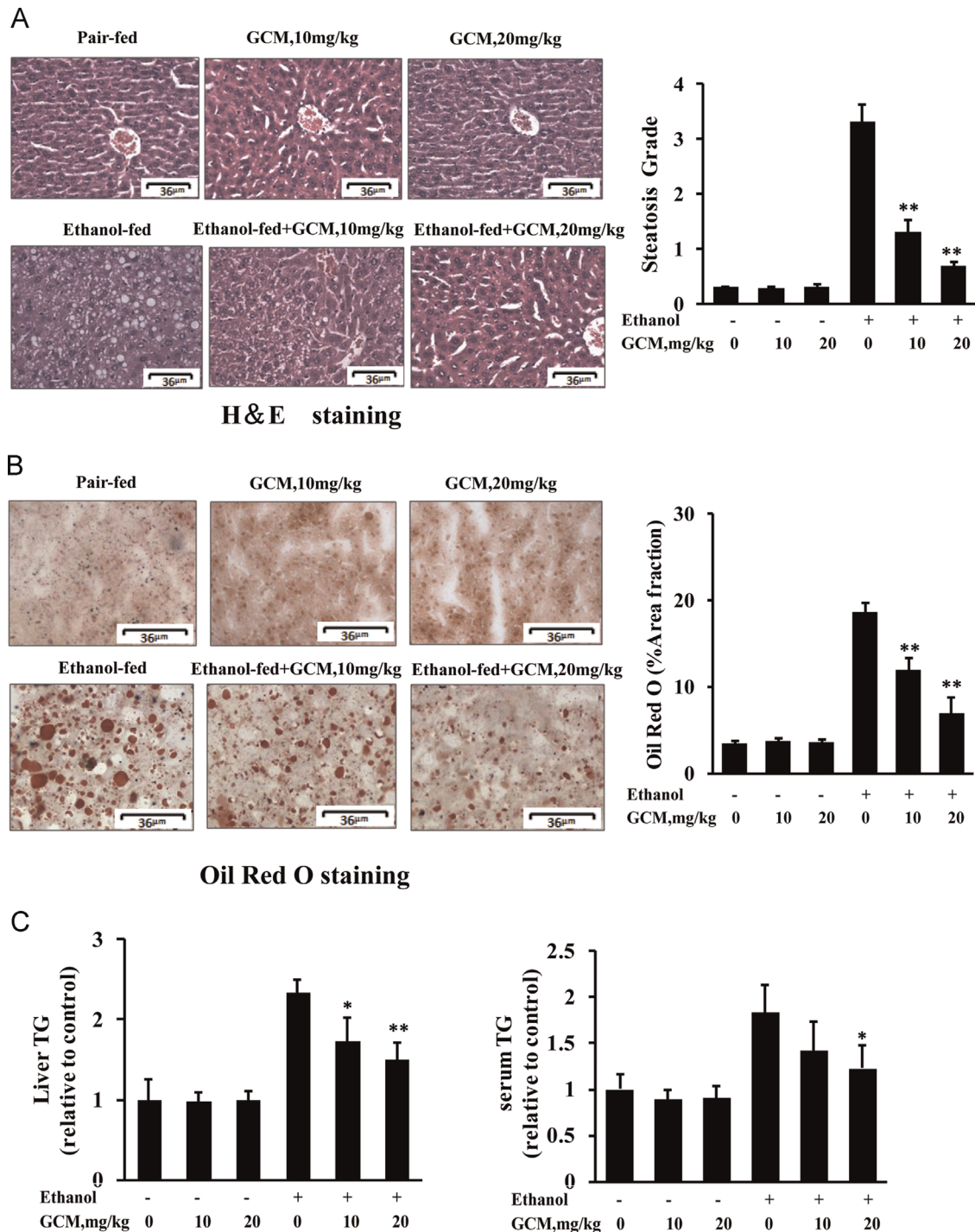


Fig. 1. GCM prevents hepatic steatosis induced by chronic ethanol exposure in vivo. Animals and treatments are as described under Materials and methods. (A) (Left) GCM protects against hepatic steatosis induced by chronic exposure as assessed by H&E staining of sections of liver tissues. Scale bars, 36 μ m. (Right) Steatosis in liver samples was graded by a blinded pathologist. (B) (Left) GCM reduces hepatic lipid storage by ethanol exposure measured by oil red O staining. Representative photomicrographs are shown depicting oil red O staining of liver sections. (Right) Quantification of oil red O-positive tissue areas was performed using ImageJ software. Values were calculated as average positive area fraction. (C) GCM decreases accumulation of triglyceride in liver and blood. Hepatic (left) and serum (right) TG concentrations were determined 8 h after ethanol treatment ($n=6$, * $P < 0.05$, ** $P < 0.01$ compared with ethanol group).

2. Results

2.1. GCM prevents hepatic steatosis induced by chronic ethanol exposure in vivo

To investigate the hepatoprotective effect of GCM, we first assessed the in vivo efficacy of GCM in a chronic alcoholic liver injury model described under Materials and methods. As shown in Fig. 1A, chronic alcohol exposure for 5 weeks resulted in significant hepatic steatosis, whereas GCM treatment alone (10 or 20 mg/kg, daily ip injection) did not cause detectable liver injury. In contrast, the steatosis induced by chronic alcohol exposure was obviously reduced by the GCM/alcohol cotreatment (analyzed by Dr. Fei Pei, a pathologist in the Department of Pathology at Peking University). Consistent with the pathogenic observations, treatment with GCM significantly decreased lipid accumulation in both

liver and blood measured by oil red O staining (Fig. 1B) or a triglyceride quantification kit (Fig. 1C). Together, these data suggest that GCM is able to effectively protect against ethanol-induced hepatic steatosis in an animal model.

2.2. GCM protects against liver damage induced by either chronic or acute ethanol exposure in vivo

To determine the protective effect of GCM on liver injury in response to chronic ethanol exposure, we measured the levels of ALT and AST, the key biochemical markers of liver injury. As shown in Figs. 2A and B, chronic ethanol exposure caused a significant increase in ALT/AST level, whereas GCM treatment alone did not change the levels of ALT/AST. The levels of ALT/AST increased by alcohol were dramatically reduced by the cotreatment of GCM and alcohol. Alcoholic liver injury may take the chronic form or that of

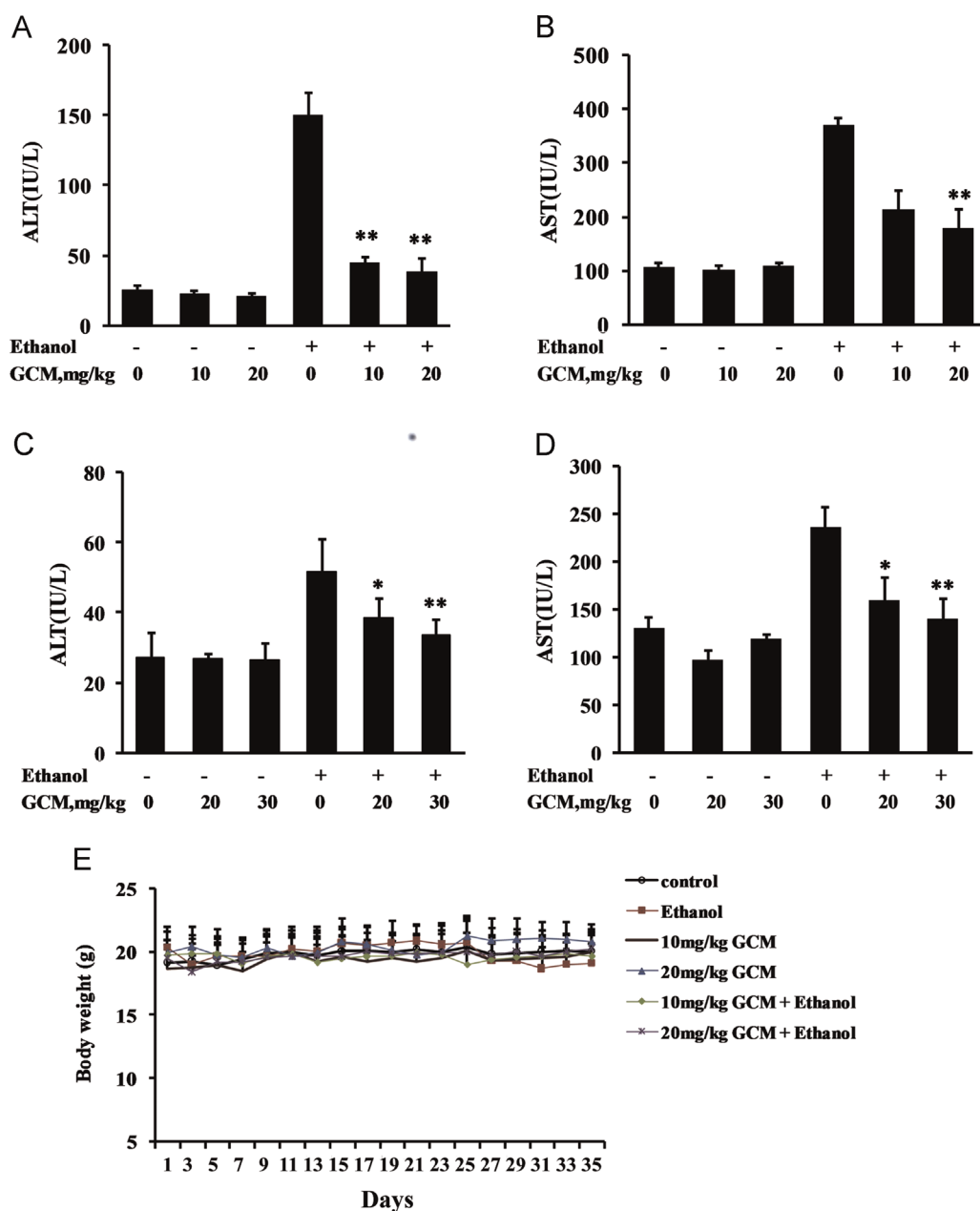


Fig. 2. GCM protects against liver damage induced by either chronic or acute ethanol exposure in vivo. (A and B) Liver injury by chronic ethanol exposure was reduced in hepatocytes of mice fed GCM. Serum ALT (A) and AST (B) were measured as described under Materials and methods. (C and D) Liver injury by acute ethanol exposure was decreased in response to GCM. Liver damage was assessed by measuring serum ALT (C) and AST (D) levels for the acute model of ethanol exposure ($n=6$, $*P < 0.05$, $**P < 0.01$ compared with ethanol group). (E) Effects of GCM and ethanol on body weight.

acute injury. Having found that GCM reduces hepatotoxicity induced by chronic alcohol exposure, we next asked if GCM can also offer protection against liver injury induced by acute alcohol exposure that is comparable with human binge drinking. As shown in Figs. 2C and D, a significantly decreased ALT/AST level was detected in the GCM/acute ethanol treatment groups in comparison

with the acute ethanol alone exposure. The alcohol-treated mice lost weight during the experiment, but there was no significant difference in the mean body weight among the groups at the end of the experiment (Fig. 2E). The data indicate that GCM can exert hepatoprotective effects in both chronic and acute alcoholic liver injury animal models.

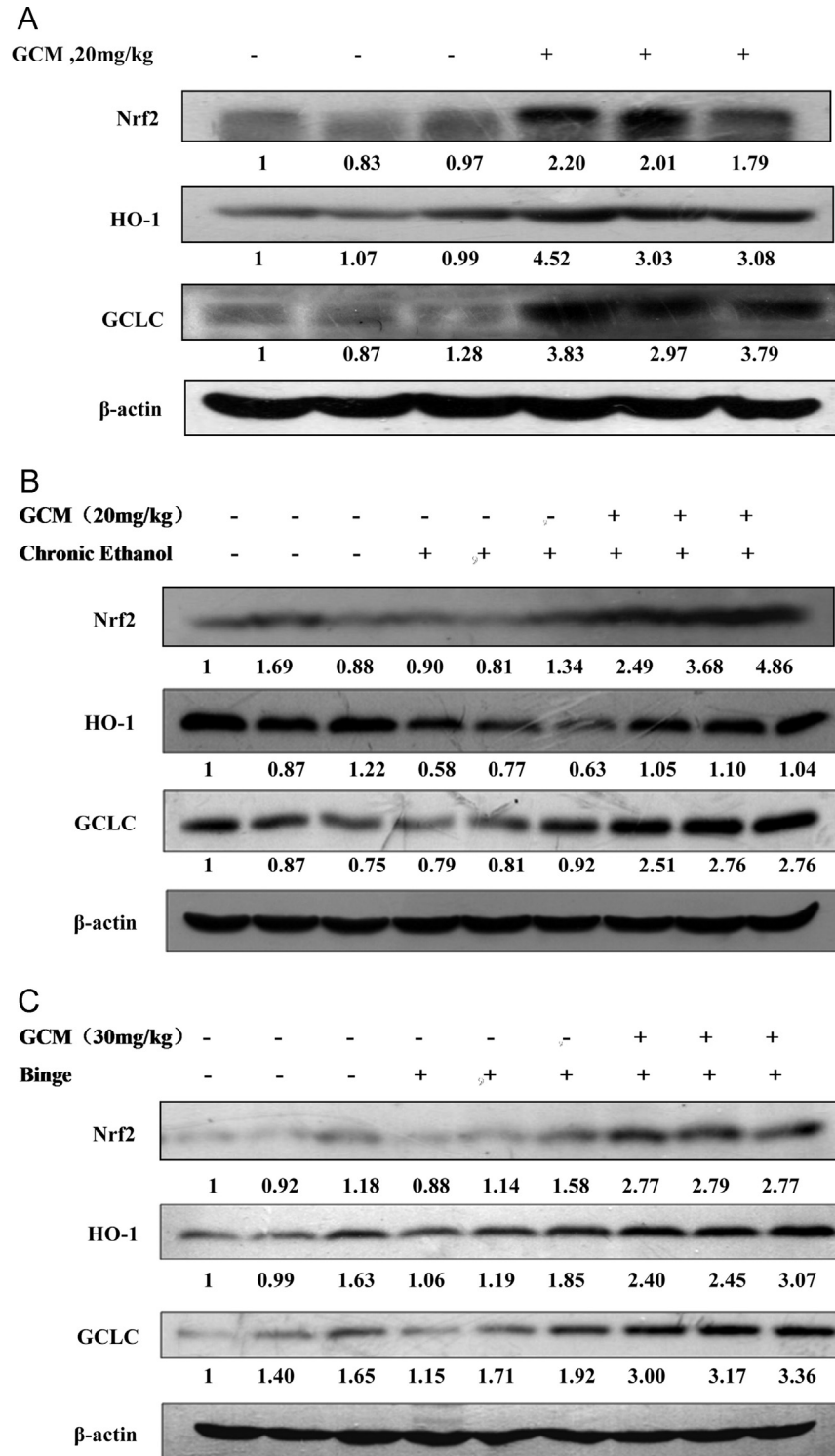


Fig. 3. Nrf2 activation contributes to the hepatoprotective effect of GCM in vivo. Animals and treatments are as described under Materials and methods. (A) Western blotting analysis of the expression of Nrf2 and the two transcriptional targets of Nrf2, HO-1 and GCLC, in response to GCM alone in vivo. (B and C) GCM treatment increased Nrf2, HO-1, and GCLC expression in the chronic (B) and acute (C) alcoholic liver injury models as analyzed by Western blotting. Relative protein levels were quantitated and indicated below each band.

2.3. Nrf2 activation by GCM in vivo

Nrf2 is a transcription factor that regulates the expression of genes involved in the cellular defense system. It has been shown that activation of Nrf2 plays a critical role in protection against alcohol-induced liver injury [16–18]. We hypothesized that regulation of the Nrf2 pathway might be involved in the hepatoprotective effect of GCM. Nrf2 and its two transcriptional targets HO-1 and GCLC were examined by Western blotting. As shown in Fig. 3A, basal levels of Nrf2, HO-1, and GCLC were apparently up-regulated in response to GCM. In the chronic ethanol exposure experiment, a slight reduction in Nrf2, HO-1, and GCLC expression

was found in ethanol treatment, whereas GCM/ethanol treatment induced a significant increase in Nrf2 and these two Nrf2 transcriptionally regulated proteins in liver tissues compared with that found in the ethanol group (Fig. 3B). In the acute ethanol exposure experiment, ethanol treatment induced a slight increase in Nrf2, HO-1, and GCLC expression, whereas GCM/ethanol treatment led to a further increase in Nrf2 and its two transcriptional targets in liver tissues (Fig. 3C). Taken together, these results indicate that treatment with GCM resulted in upregulation of the Nrf2 downstream targets HO-1 and GCLC in both chronic and acute alcohol-exposure models, suggesting that Nrf2 was transcriptionally activated by GCM exposure in vivo.

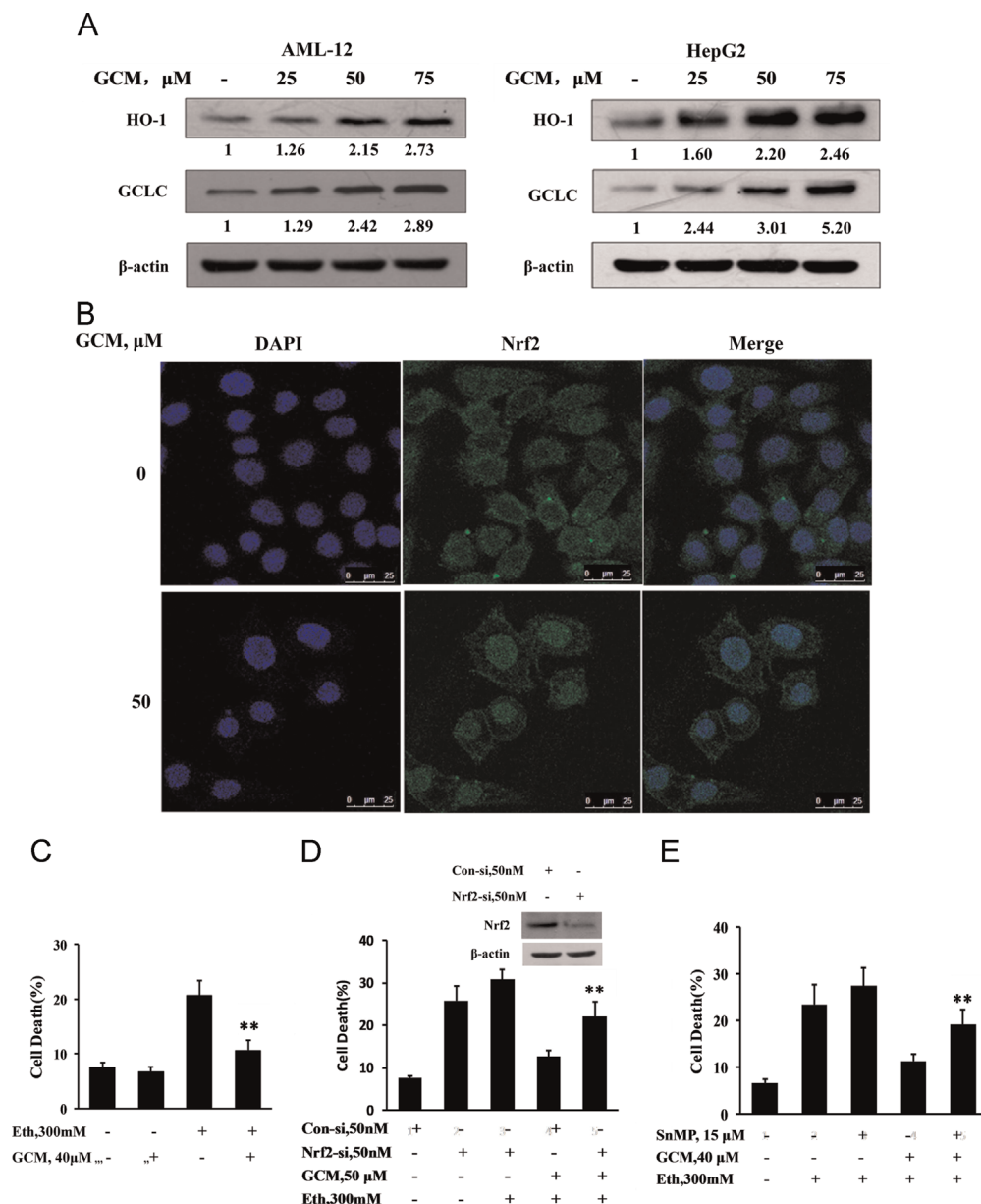


Fig. 4. GCM activates Nrf2 in a cell culture model. (A) GCM upregulates Nrf2 transcriptional targets HO-1 and GCLC in AML-12 and HepG2 cells. The cells were treated with various concentrations of GCM for 24 h and expression of HO-1 and GCLC was measured by Western blotting. Relative protein levels were quantitated and indicated below each band. (B) GCM induces Nrf2 nuclear translocation in HepG2 cells. The cells were treated with 50 μ M GCM for 24 h and the localization of Nrf2 was detected by immunofluorescence staining. Scale bars, 25 μ m. (C) GCM protects against ethanol-induced cytotoxicity in AML-12 cells. The cells were treated with ethanol in the presence or absence of GCM for 24 h and cell death was measured by annexin V staining (** P < 0.01 compared with Eth group). (D) Influence of Nrf2 knockdown on the protective effect of GCM against ethanol-induced cell death measured by annexin V staining (** P < 0.01 compared with Con-si-, ethanol-, and GCM-treated cells). (E) Influence of HO-1 inhibition by its inhibitor SnMP on the protective effect of GCM against ethanol-induced cell death measured by annexin V staining (** P < 0.01 compared with ethanol- and GCM-treated cells).

2.4. GCM ameliorates ethanol-induced cytotoxicity via Nrf2 activation in vitro

Having found Nrf2 activation in response to GCM exposure, we next asked whether Nrf2 activation indeed played a functional role in the hepatoprotective effect of GCM and how GCM triggers Nrf2

activation. To address these issues, a suitable in vitro model was needed. For this purpose, AML-12, a mouse hepatocyte cell line, and HepG2, a human liver carcinoma-derived cell line, were tested. The cells were treated with increasing concentrations of GCM for 24 h and Western blotting was used to examine the changes in Nrf2-regulated proteins. As shown in Fig. 4A, HO-1 and GCLC, the two

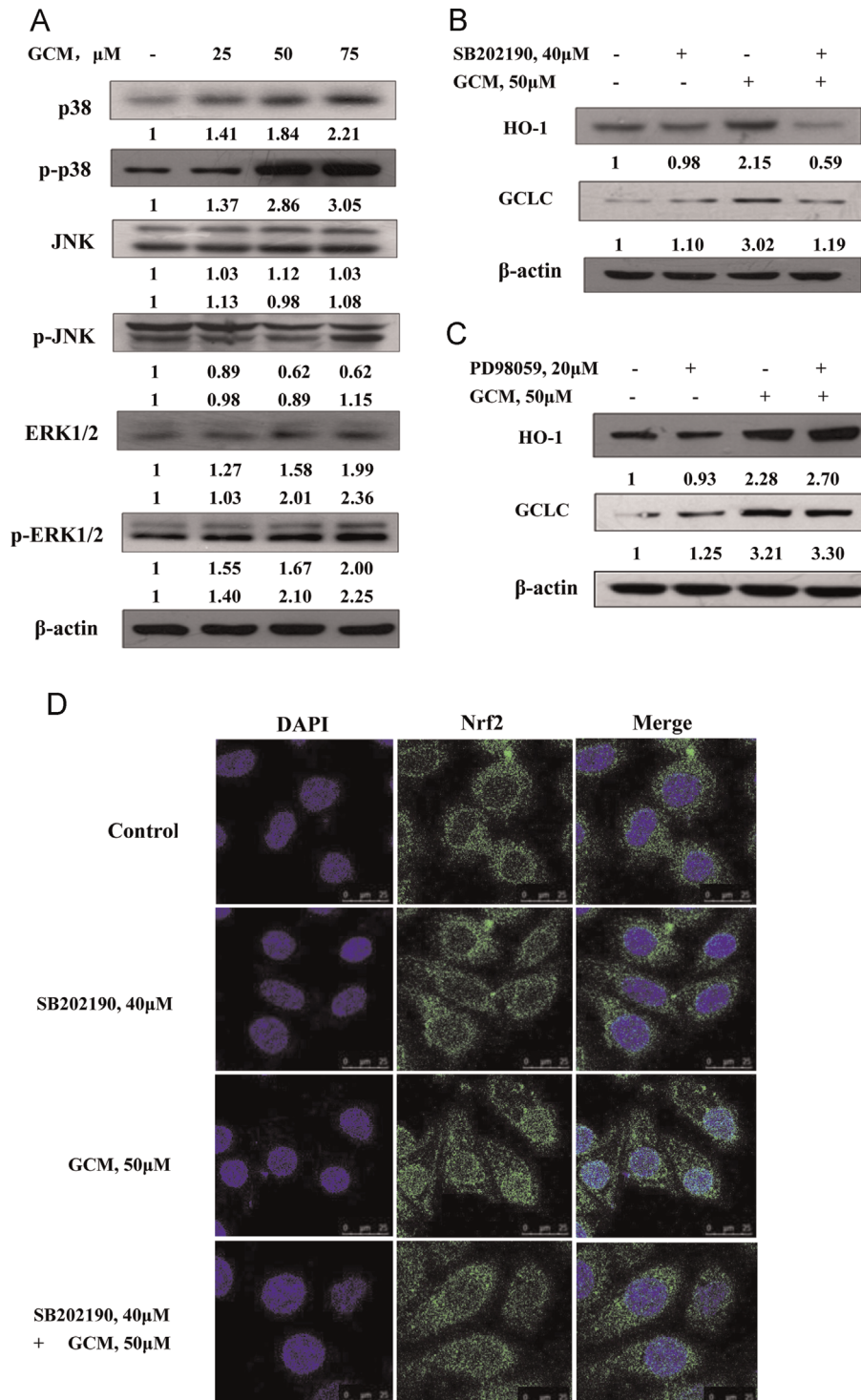


Fig. 5. p38 activation is involved in GCM-induced Nrf2 activation. (A) Effects of GCM treatment on MAPKs. The cells were treated with various concentrations of GCM for 24 h. The total and the phosphorylated p38, JNK1/2, and ERK1/2 were analyzed by Western blotting. (B) Effects of p38 inhibitor on GCM-induced changes in Nrf2 target genes. The cells were treated with 50 μM GCM in the presence or absence of SB202190 for 24 h, and the Nrf2 target genes GCLC and HO-1 were analyzed by Western blotting. (C) Effects of ERK1/2 inhibitor on GCM-induced changes in Nrf2 target genes. The cells were treated with 50 μM GCM in the presence or absence of PD98059 for 24 h, and the Nrf2 target genes GCLC and HO-1 were analyzed by Western blotting. Relative protein levels were quantitated and indicated below each band. (D) Effects of p38 inhibitor on GCM-induced Nrf2 nuclear translocation analyzed by immunofluorescence staining.

transcriptional targets of Nrf2, were induced by GCM in both cell lines tested. This result was further validated by immunofluorescence staining of Nrf2 localization. As shown in Fig. 4B, Nrf2 was localized primarily in the cytoplasm in the untreated cells, whereas GCM treatment significantly increased the nuclear translocation of Nrf2, further supporting activation of Nrf2 in response to GCM in the in vitro cell culture model. Moreover, we found that GCM was able to significantly protect against ethanol-induced

cytotoxicity in AML-12 cells (Fig. 4C). Regarding the functional role of Nrf2 activation in the cell culture model, we tested the influence of Nrf2 or HO-1 inactivation on the protective effect of GCM against ethanol-induced cytotoxicity. As shown in Fig. 4D, under conditions in which Nrf2 was silenced by RNA interference, the protective effect of GCM on ethanol-induced cell death was significantly attenuated. Similar results were also obtained when HO-1 was inhibited by its inhibitor stannic mesoporphyrin (SnMP) (Fig. 4E).

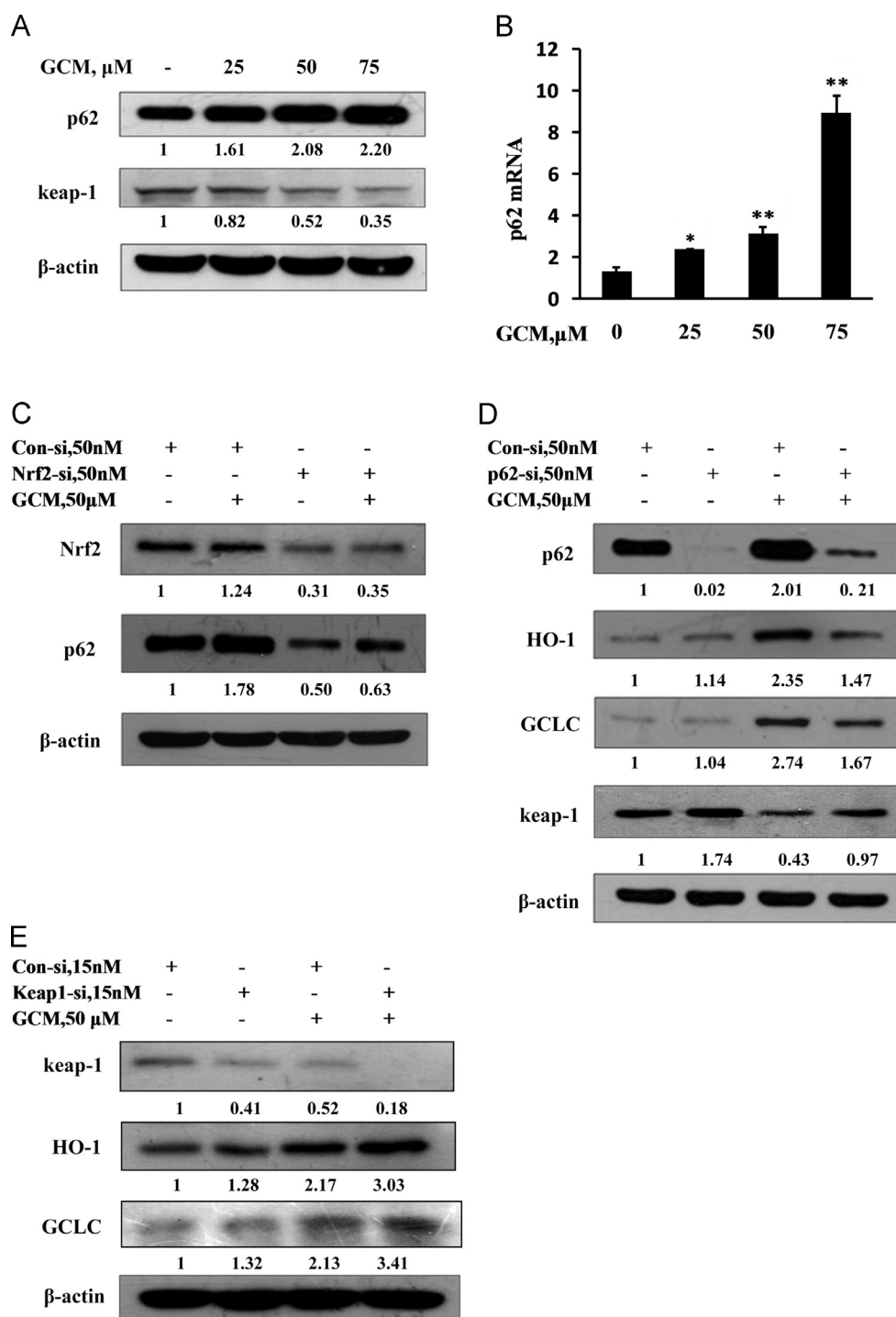


Fig. 6. Transcriptional upregulation of p62 contributes to Nrf2 activation. (A and B) Effects of GCM treatments on p62 and Keap1 expression. The cells were treated with various concentrations of GCM for 24 h. The protein levels of p62 and Keap1 and mRNA level of p62 were analyzed by Western blotting (A) and RT-PCR (B), respectively (* P < 0.05, ** P < 0.01 compared with GCM-untreated cells). (C) Effects of Nrf2 inactivation by RNA interference on GCM-induced p62 induction analyzed by Western blotting. (D) Effects of p62 inactivation by RNA interference on GCM-induced Nrf2 target genes and Keap1 expression analyzed by Western blotting. Relative protein levels were quantitated and indicated below each band. (E) Effects of Keap1 inactivation by RNA interference on GCM-induced Nrf2 target genes expression analyzed by Western blotting.

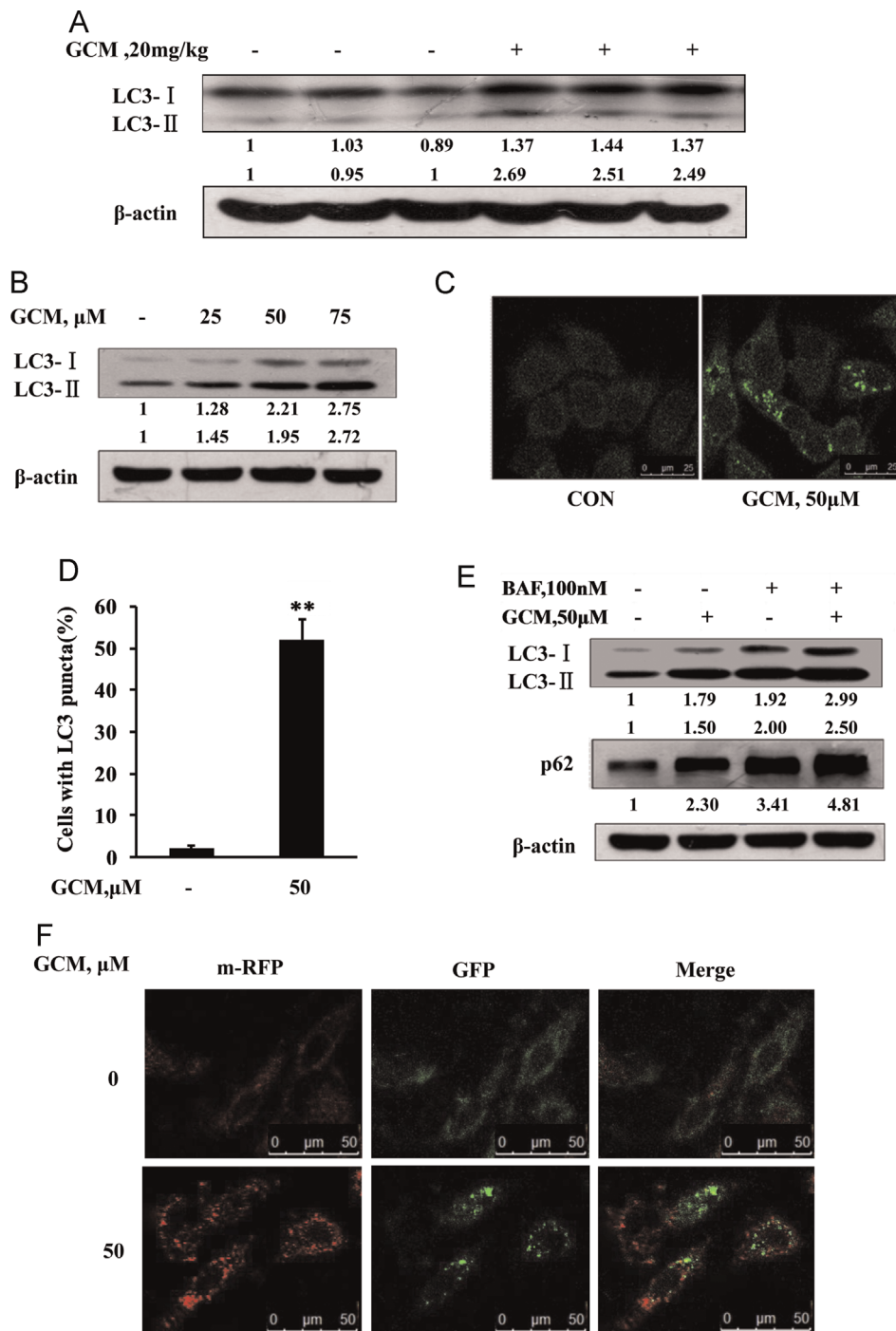


Fig. 7. GCM induces autophagy in HepG2 cells. (A) Protein expression detected by Western blotting in vivo. (B and C) GCM increases LC3-I to LC3-II conversion and induces punctate distribution of LC3. The cells were treated with 50 μ M GCM for 24 h and then LC3-I to LC3-II conversion was analyzed by Western blotting (B) and punctate distribution of LC3 was detected by immunofluorescence staining (C). Scale bars, 25 μ m. (D) Quantitative analysis of LC3-punctate cells in GCM-treated and untreated cells (** $P < 0.01$ compared with GCM-untreated cells). (E) Increase in autophagosome formation was involved in GCM-induced autophagy in HepG2 cells. The cells were treated with 25 μ M GCM for 24 h in the presence or absence of 50 nM bafilomycin A1 (added 2 h before cell harvest) and then LC3 and p62 were analyzed by Western blotting. Relative protein levels were quantitated and indicated below each band. (F) Monitoring autophagic flux by mRFP–GFP–LC3 reporter.

Together, these results support a critical role for Nrf2 activation in GCM-mediated protective effects against ethanol-induced hepatotoxicity.

2.5. p38 activation is involved in GCM-induced Nrf2 activation

Having established the role of Nrf2 activation in hepatoprotective activity of GCM, we next investigated the mechanisms involved in GCM-induced Nrf2 activation. Among various upstream

mediators of Nrf2 activation, mitogen-activated protein kinases (MAPKs) are suggested to play a critical role in some model systems [19–21]. We next asked if MAPKs are the upstream kinases involved in GCM-induced Nrf2 activation. Total and phosphorylation status of the three key MAPK members, ERK1/2, p38, and JNK1/2, was analyzed by Western blotting. As shown in Fig. 5A, GCM treatment induced a concentration-dependent increased expression and phosphorylation of p38 and ERK1/2, but not JNK1/2. To determine the functional role of p38 and ERK1/2

phosphorylation, we tested the influence of their inhibitors on GCM-induced upregulation of the Nrf2 targets HO-1 and GCLC. The cells were treated with 50 μ M GCM in the presence or absence of SB202190 or PD98059 for 24 h and changes in HO-1 and GCLC expression were assessed by Western blotting. Under the condition in which p38 (Fig. 5B) but not ERK1/2 (Fig. 5C) was inhibited, GCM-induced upregulation of HO-1 and GCLC was significantly reduced. Furthermore, the effect of p38 inactivation on Nrf2 activation induced by GCM was confirmed by analyzing its nuclear translocation. As shown in Fig. 5D, when p38 was blocked by its inhibitor, GCM-induced Nrf2 nuclear translocation was significantly attenuated. Together, these results suggest that p38 but not ERK1/2 activation lies upstream of Nrf2 activation by GCM.

2.6. Transcriptional upregulation of p62 contributes to Nrf2 activation

It has been shown that p62 is a transcriptional target of Nrf2 [22,23] and p62 accumulation is associated with liver disease in certain situations [24]. Having found activation of Nrf2 by GCM, we then questioned if p62 was transcriptionally upregulated in response to GCM. If it is, does p62 upregulation play any role in the current system? We first measured the levels of p62 protein and mRNA in response to GCM and the results are shown in Figs. 6A and B. Both protein and mRNA levels of p62 increased, suggesting p62 was transcriptionally induced by GCM. To determine the role of Nrf2 in induction of p62 by GCM, we assessed the effect of Nrf2 knockdown on GCM-induced p62 expression. As shown in Fig. 6C, under conditions of Nrf2 silencing, upregulation of p62 by GCM was dramatically reduced, indicating a critical role for Nrf2 activation in GCM-induced p62 upregulation. It has been shown that p62 can interact with Keap1, an adaptor of the Cul3-ubiquitin E3 ligase complex responsible for degrading Nrf2, leading to inhibition of Keap1-dependent Nrf2 degradation [25]. We then asked if p62 induction contributed to Nrf2 activation by GCM via suppression of Keap1. Western blot analysis of Keap1 showed that GCM caused a concentration-dependent downregulation of Keap1 expression (Fig. 6A). Under conditions of p62 silencing, downregulation of Keap1 by GCM was nearly abolished, whereas GCM-induced HO-1 and GCLC were significantly decreased (Fig. 6D). To critically test the role of Keap1 in GCM-induced Nrf2 activation, we examined the influence of Keap1 inhibition by an RNAi approach on Nrf2 activation. As shown in Fig. 6E, knockdown of Keap1 led to increased expression of HO-1 and GCLC, whereas GCM treatment caused an inhibition of Keap1 expression comparable to that with the knockdown approach and resulted in a stronger induction of HO-1 and GCLC compared with Keap1 knockdown. Under the condition of Keap1 knockdown, a further increased expression of HO-1 and GCLC was observed in response to GCM exposure. Taken together, these results suggest that Nrf2 activation leads to transcriptional upregulation of p62, which in turn contributes to GCM-induced Nrf2 activation through suppression of Keap1, which forms a positive feedback loop for Nrf2 activation.

2.7. GCM activates autophagy

It has been shown that autophagy induction can reduce alcohol-induced hepatotoxicity possibly associated with removing damaged mitochondria and accumulated lipid droplets [26–28]. We next asked if autophagy was also involved in hepatoprotective activity of GCM. A cleaved and lipidated form of LC3, LC3-II, an autophagic marker, was analyzed by Western blotting. As shown in Fig. 7A, treatment with GCM at concentrations of 25 to 75 μ M for 24 h led to a significant increase in LC3-II level in HepG2 cells. LC3 lipidation by GCM was further verified by immunofluorescence staining to analyze distribution patterns of LC3. As

shown in Fig. 7B, a diffuse localization of LC3 fluorescence was observed in untreated cells, whereas a punctate pattern of LC3 fluorescence was observed in GCM-treated cells. Quantitative analysis of LC3-puncta-positive cells showed that the number of cells with LC3 puncta in GCM-treated cells was significantly increased compared with the untreated control (Fig. 7C). We next measured autophagic flux to determine whether the elevated levels of LC3 cleavage and lipidation induced by GCM were due to increased formation of autophagosomes or to decreased degradation. As shown in Fig. 7D, treatment with bafilomycin A1 (BAF), an inhibitor of autophagosome degradation, was expected to increase accumulation of LC3-II, whereas GCM treatment significantly increased LC3-II levels in either the presence or the absence of BAF. Similar changes in p62 were also found under such conditions, suggesting the autophagic flux is increased in response to GCM. This result was further validated by an immunofluorescence approach using the mRFP–GFP–LC3 reporter [29]. As shown in Fig. 7E, both early (green/red colocalization) and late (red alone) autophagic vacuoles were increased in response to GCM. Furthermore, an increased LC3-II level by GCM was observed in vivo (Fig. 7F). Together, the data suggest that autophagy is activated by GCM and is probably also attributable to the GCM-mediated hepatoprotective effect.

3. Discussion

In clinical and animal studies, licorice has been shown to have multiple biological activities, including a hepatoprotective effect. Although the chemistry of licorice has been extensively studied, the active components that contribute to its biological functions are far from clear. Identification of active ingredients from licorice is needed for a better utilization of this popular edible and medicinal plant. It has been documented that the main bioactive chemical constituents in licorice include flavonoids, triterpene saponins, and coumarins [8]. GCM is a major bioactive coumarin compound isolated from licorice and the hepatoprotective activity of GCM has not been scientifically addressed. In the current study, we have tested the hepatoprotective effect of GCM using animal models of both chronic and acute alcoholic liver injury and found for the first time that GCM possesses a potent protective effect against ethanol-induced hepatotoxicity in both experimental animal models through mechanisms involved in Nrf2 activation and autophagy induction. Our findings support GCM as a novel active compound that is responsible for the hepatoprotective activity of licorice and provide a novel mechanistic support for a hepatoprotective function of licorice.

Nrf2 is a key player in the cellular antioxidant defense system. The protective role of Nrf2 in alcohol-induced hepatotoxicity is well established [16–18]. Regarding the mechanisms underlying the hepatoprotective effect of GCM, our data showed that GCM induced Nrf2 activation both in vivo and in vitro. Under conditions of Nrf2 deficiency, GCM failed to exert the protective effect on alcohol-induced liver injury in the Nrf2-knockout mouse model, clearly supporting involvement of an Nrf2 pathway in the hepatoprotective effect of GCM. We further delineated the mechanisms of Nrf2 activation by GCM using a cell culture model and demonstrated both p38- and p62-dependent mechanisms contributing to GCM-induced Nrf2 activation.

p62, also known as sequestosome-1, is a substrate of autophagy [30]. The biological functions of p62 remain controversial. Under normal conditions, p62 can recognize toxic cellular waste (toxic aggregate-prone proteins, invading microorganisms, and damaged organelles), which is then cleared by autophagy. Under the condition that autophagy is disrupted, p62 accumulates with aggregates of ubiquitinated proteins, which in turn leads to several

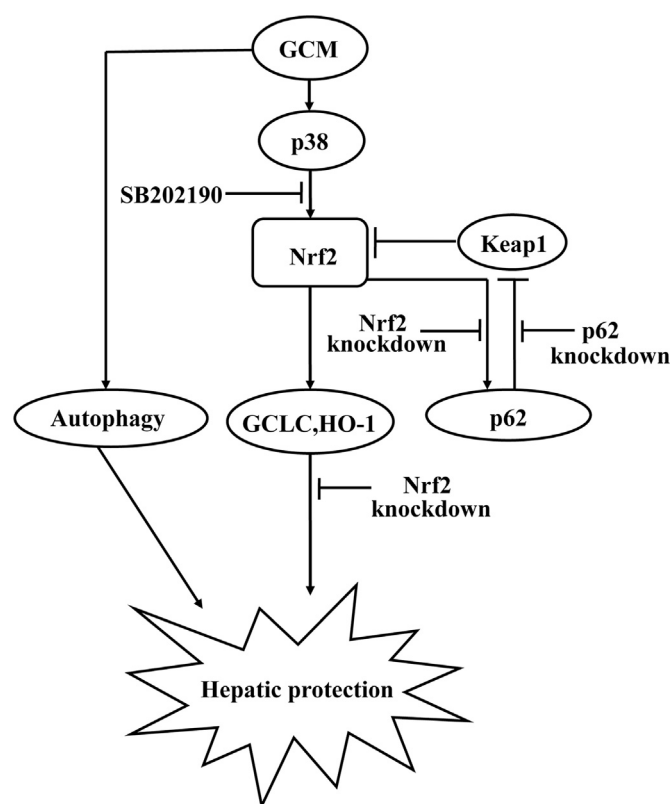


Fig. 8. Signaling pathways underlying GCM-induced Nrf2-mediated preventive effects on alcohol-induced hepatotoxicity. GCM activates p38, which in turn leads to activation of Nrf2. Transcriptional upregulation of p62 by Nrf2 promotes autophagic degradation of Keap1, which further contributes to Nrf2 activation.

pathological consequences including liver disease [31–33]. The pathological consequences of p62 accumulation by autophagy deficiency have been well documented. However, less is known about the biological functions of p62 induction in liver disease under conditions of autophagy sufficiency. Keap1 has been identified as a binding partner of p62 and the direct interaction between p62 and Keap1 disrupts the association of Keap1 and Nrf2, leading to stabilization and transcriptional activation of Nrf2 [22,34,35]. Inhibition of Keap1 by p62 is therefore considered to be an important mechanism contributing to regulation of Nrf2 activation. Regarding the mechanisms of Nrf2 activation in response to GCM, we evaluated the role of p62 in the regulation of the Keap1–Nrf2 system. The results indicated that autophagy was induced rather than inhibited by the treatment and p62 was upregulated through a transcriptional mechanism rather than degradation inhibition by autophagy deficiency. When p62 was silenced, GCM-induced Keap1 degradation was almost abolished and Nrf2 activation by GCM was dramatically reduced, indicating involvement of p62 induction in Nrf2 activation by GCM. Given the pivotal role of Nrf2 in protection against ALD, these data clearly suggest that p62 induction may be beneficial in preventing and protecting against alcohol-induced liver injury under the current conditions (autophagy sufficiency), which is possibly associated with Keap1 degradation-dependent activation of Nrf2.

Modulation of autophagy was considered to be an effective approach to alleviate liver injury [26–28]. In the current study we have evaluated the effect of GCM on autophagy induction and showed that GCM caused a significant induction of autophagy. We also found that autophagy inhibition abolished GCM-induced Keap1 downregulation (data not shown), suggesting autophagy induction by GCM is required for p62-mediated Keap1 degradation. Transcriptional upregulation of p62 worked together with

autophagy induction in promoting Keap1 degradation, which in turn contributed to Nrf2 activation in our current system.

4. Conclusions

In summary (Fig. 8), we have identified GCM, a major bioactive coumarin of licorice, as a promising novel hepatoprotective agent. GCM exerts hepatoprotective activity through possible mechanisms involved in activation of Nrf2 and induction of autophagy. Under the condition of autophagy sufficiency, transcriptional up-regulation of p62 may offer benefits for alcohol liver via p62-dependent autophagic degradation of Keap1 and activation of Nrf2.

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